Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/06/2020
PI Name:	Laughlin, Mitzi Ph.D.		
Project Title:	Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77204-6015	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	02/23/2016	**End Date:**	08/24/2020
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 8/24/2020 per HHC element/JSC and NSSC information (Ed., 12/10/20)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Clarke, Mark SF Ph.D. (University of Houston) Layne, Charles S Ph.D. (University of Houston)		
Grant/Contract No.:	NNX16AF63G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The grant NNX16AF63G Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES began in February 2016. The study is designed to examine declines in muscular strength, function, and neuromotor control during long-duration spaceflight using the Muscle Atrophy Research and Exercise System (MARES) device in the Columbus module on the International Space Station (ISS). The knowledge gaps addressed by this study relate to the risks to human space travelers associated with in-flight changes in muscle function (PRD Risk: Risk of Impaired Performance Due to Reduced Muscle Mass, Strength, and Endurance), as identified by Integrated Research Plan (IRP) Gaps M24 (What is the time course of changes in muscle protein turnover, muscle mass and muscle function during long-duration spaceflight?) and M2 (What is the current status of in-flight/post-flight exercise performance capability?). Specifically, our study seeks to characterize adaptive in-flight changes in lower limb muscle strength and neuromotor control performance using the Muscle Atrophy Research and Exercise System (MARES).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed investigation seeks to systematically examine the relationships among muscle strength and neuro-motor control performance. By tracking these relationships over time during flight we will be able to identify any adaptive changes in integrated motor performance. If successful, this investigation should provide significant benefits to the Human Research Program by quantifying the aspects of the Program Requirements Document (PRD) Risk of Impaired Performance Due to Reduced Muscle Mass, Strength and Endurance embodied in IRP (Integrated Research Plan) Gap M24 (What is the time course of changes in muscle protein turnover, muscle mass and muscle function during long-duration spaceflight?) and M2 (What is the current status of in-flight/post-flight exercise performance capability?). Additionally, this investigation will provide valuable information to clinicians regarding muscle atrophy due to injury or disease on Earth.
Task Progress:	Scientific Report Abstract Introduction: Spaceflight induces significant loss of skeletal muscle mass and function in response to mechanical unloading of the neuromusculoskeletal system. As the means to assess strength during flight have not routinely been available, most of the data used to formulate these hypotheses have come from post-flight assessments (relative to pre-flight values). Thus the purpose of this study is to examine the effects of spaceflight on knee and ankle strength and neuromuscular activation during 6 months of spaceflight. Methods: Four crewmembers were tested on board the International Space Station (ISS) after 8.9±3.0, 49.1±10.1, and 131.5±27.7 days in-flight, as well as pre- and post-flight, using the Muscle Atrophy Research and Exercise System (MARES). Voluntary isometric torque during maximal knee extension and flexion and ankle plantarflexion. The knee was tested isometrically at 90, 60, and 45 degrees of flexion and isokinetically at 60, 120, 180, 240, and 300 degrees per second. The ankle plantarflexors were tested at 20, 10, 0, and -10 degrees for isometric testing and 30, 50, 100, and 150 degrees per second for isokinetic testing. Each angle and speed was tested in duplicate and the highest torque produced was used for analysis. Surface EMG was simultaneously measured during all tests. Root mean squared (RMS) EMG within a 500 ms window centered on the time of peak torque was identified, and ratios of EMG to peak torque (EMG/T) were calculated for each contraction. Results: Peak knee extension and flexion strength declined immediately inflight from preflight or baseline values (extension -24% to -36% and flexion -26% to -31%) and both showed a gradual but not full recovery by the third inflight test session. Ankle plantarflexion strength was maintained at the first inflight session but declined at the second test session. There was a slight recovery prior to landing but deficits were 4-18% as compared to baseline measures. On landing, ankle plantarflexor muscle strength was similar to the final inflight test session which were lower than baseline measures. Neuromusclar activation during flight as measured by EMG generally showed that greater muscle activation was needed to produce similar levels of torque for both the knee and ankle testing sessions. Recovery to baseline strength after landing was gradual and was not complete at R+60. The recovery curve was individualized but showed a pattern of decline at R+6 and R+15 with a gradual increase to R+30 and R+60. No crewmember showed full recovery at R+60 for both knee and ankle strength testing. Neuromuscular activation during the recovery period mostly returned back to baseline levels meaning that the muscle activation required to produce a level of torque was similar at baseline and R+60; however ,these trends should be used cautiously as EMG data was frequently lost due to equipment issues. Conclusions: Results suggest that spaceflight results in an immediate decline of peak torque production for both isometric and isokinetic knee extension and flexion while ankle plantarflexion is somewhat protected until midflight. Current in-flight countermeasures appear to trend towards functional restoration of knee extension while knee flexion remained relatively unchanged and ankle plantarflexion strength was mixed. Upon landing, ankle plantarflexors maintained strength about equal to the last inflight measure. Muscle strength during the recovery period declined after the landing measure (R+6 and R+15) and then gradually improved at R+30 and R+60. Although much improved, no crewmember returned to 100% for all three muscle groups by the final test.
Bibliography Type:	Description: (Last Updated: 12/10/2020)
Abstracts for Journals and Proceedings	Reed JT, Layne CS, Koryak Y, Clarke MSF, Tomilovskaya E, Albracht K, Shenkman B, Stäudle B, Rittweger J, Kozlovskaya I, Laughlin MS. "SarcoLab-3: Changes in Knee Flexor and Extensor Torque Generation during a Six-Month Space Flight Mission." Presented at the American College of Sports Medicine 66th Annual Meeting, Orlando, FL, May 28-June 1, 2019. Medicine & Science in Sports & Exercise. 2019 Jun;51(6 Suppl):407. , Jun-2019
Abstracts for Journals and Proceedings	Laughlin MS, Reed JT, Layne CS, Koryak Y, Clarke MSF, Tomilovskaya E, Albracht K, Shenkman B, Stäudle B, Kozlovskaya I, Rittweger J. "SarcoLab-3: Changes in Knee Flexor and Extensor Torque Generation during a Six-Month Space Flight Mission." 2020 NASA Human Research Program Investigators' Workshop – Human Exploration Small Steps Lead to Giant Leaps: Translating Research into Space Exploration, Galveston, Texas, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 27-30, 2020. , Jan-2020